

GiniDispatch: A GINI-REGULARIZED REINFORCEMENT LEARNING FOR JOINT EARNINGS AND WORKLOAD EQUITY IN TAXI DISPATCH

Abstract

Ride-hailing platforms typically optimize dispatch decisions for passenger waiting time and platform revenue, yet these objectives often produce persistent inequalities in driver income and workload. This study introduces GiniDispatch, a fairness-aware taxi dispatch algorithm that improves driver welfare while maintaining operational performance. The framework integrates three components: fairness-prioritized ordering of ride requests, equity-driven driver-request matching, and a reward structure penalizing income and workload inequality using Gini-based measures. GiniDispatch is implemented with two reinforcement learning approaches—Spatiotemporal Graph Convolutional Reinforcement Learning (STGCRL) and tabular Q-Learning—and evaluated across varying demand conditions and fairness-weighting settings using New York City taxi trip data. Results show that GiniDispatch reduces income inequality by up to 34.3% and workload inequality by up to 40.1% compared to fairness-unaware baselines. Configurations balancing income and utilization objectives consistently yield the highest overall driver welfare. These equity improvements are achieved with minimal changes in passenger waiting time and total reward, suggesting that incorporating fairness considerations into dispatch decisions does not necessarily compromise operational efficiency. Although STGCRL produces the largest equity improvements, simpler agents also yield measurable gains, indicating that equity-enhancing dispatch mechanisms can be implemented without highly complex system architectures. Overall, the results provide practical insights into how operational objectives may be balanced with considerations of driver welfare.

Keyword: Taxi Dispatching, Fairness-aware Taxi Dispatching, Reinforcement Learning,

1. Introduction

Taxi dispatching (TD) is a crucial component of urban mobility platforms. It involves the real-time bipartite matching of available taxis with pending requests, coordinated through a digital platform (Sühr et al., 2019). Traditionally, the TD algorithms have been designed to optimize operational efficiency, focusing on metrics involving profitability and passenger satisfaction (Xu & Xu, 2020; Brar & Su, 2020). However, this efficiency-oriented approach more often results in large earning disparities among drivers (Bokányi & Hannák, 2019; Sun et al., 2022; Dang et al., 2025), inflicting envy among peer drivers (Aleksandrov, 2023). Such perceptions of envy can negatively affect system performance, as dissatisfied drivers are more likely to cancel requests, leading to a rise in unfulfilled trips and longer wait times for passengers (Graham & Joe, 2017; Nanda et al., 2020). From the driver's perspective, this behavior is understandable, as many struggle to earn more than minimum wage—or even avoid negative income, where operational expenses exceed earnings (Graham & Joe, 2017; Zoepf et al., 2018). Accordingly, designing a fair TD system that incorporates driver equity is essential for enhancing both operational efficiency and social welfare within the urban mobility platforms.

From a social welfare perspective, fairness emerges as a critical objective for TD systems. Promoting income equality helps ensure that drivers are compensated in proportion to the time they spend on the platform (Sühr et al., 2019). This, in turn, fosters a perception of fairness among drivers, increasing their likelihood of complying with system recommendations (Brar & Su, 2020). Prior studies have demonstrated that fair TD systems can significantly enhance both operational efficiency and driver satisfaction (Xu & Xu, 2020; Xie et al., 2023). However, fairness-oriented

research often overlooks another key dimension: workload equality. For instance, it is also unfair when drivers must work significantly longer hours to achieve the same income as others. Although many drivers willingly extend their hours to meet earnings goals, research has shown that most would prefer to reduce their working time if they could meet those goals more efficiently (Allon et al., 2023). Moreover, an expanding body of research highlights the adverse effects of extended work hours on taxi drivers, including increased fatigue (Lim & Chia, 2015), job stress (Göktaş, 2023), and a higher risk of traffic accidents (Meng et al., 2015). Therefore, fairness-aware TD systems should consider not only income equality but also equity in working hours among drivers.

Researchers have increasingly explored reinforcement learning (RL) as an alternative approach to TD, given its potential to learn adaptive ride-matching and taxi-repositioning policies from experience (Sun et al., 2022; Liang, 2024). Two key considerations in RL-based TD research are agent configuration and spatial representation. First, agent configuration can be either centralized (single-agent) or decentralized (multi-agent) (Wang et al., 2021; Rajeh et al., 2024). In centralized settings, a single agent learns a global dispatch policy using system-wide information, such as the locations of all taxis and outstanding ride requests. This approach benefits from a comprehensive view of the environment but may suffer from scalability issues and slower responsiveness in large, dynamic urban systems. In contrast, decentralized settings assign individual agents to taxis, which learn and act based on local observations or shared information. While this enhances scalability and responsiveness, it can introduce coordination challenges and suboptimal outcomes if agents act selfishly or lack sufficient communication. Secondly, the choice of spatial representation significantly impacts model performance. Traditional RL-based TD methods often rely on grid-based spatial discretization (Wang et al., 2021), which simplifies learning but may distort actual travel distances and road connectivity. More recent work leverages street network-based (graph-

based) representations to better model real-world spatial dependencies and road geometry (Kim & Kim, 2021; Pan & Liu, 2023). Graph-based representations more accurately reflect urban topology and travel dynamics, which can lead to more realistic and effective dispatch policies. Overall, while RL offers promising adaptability for TD systems, its success hinges on appropriate agent design and spatial modeling. Careful selection of these components is critical to the TD systems for balancing performance, scalability, and real-world applicability.

This paper introduces GiniDispatch, a Gini-regularized, fairness-aware centralized TD algorithm that improves driver equity while maintaining operational efficiency through a three-stage mechanism comprising fairness-aware action selection, equity-driven taxi assignment via Gini-based simulations, and fairness-augmented reward shaping. Using large-scale simulations on the NYC Taxi dataset with 250 fixed taxis under low (250), moderate (750), and high (1500) demand regimes, we first show that two reinforcement learning agents—a state-of-the-art Spatiotemporal Graph Convolutional Reinforcement Learning (STGCRL) model and a tabular Q-learning agent—consistently outperform widely used non-learning baselines, including Nearest Greedy and Hungarian matching, thereby establishing strong learning-based dispatch performance. Building on this foundation, integrating GiniDispatch further enhances equity outcomes: relative to non-fair RL baselines, GiniDispatch achieves up to 34.3% improvement in income equality and up to 40.1% improvement in workload equity, as measured by reductions in the respective Gini coefficients, across six fairness configurations. Cohen's d effect size analysis confirms that these equity gains are practically significant—with workload and earnings equality producing consistently large effects—while base reward, waiting time, and trip efficiency exhibit negligible to small effects, demonstrating that fairness-aware dispatch operates without meaningful cost to platform performance.

The main contributions of this study are as follows:

- It introduces GiniDispatch, a novel Gini-regularized, fairness-aware ride assignment algorithm for centralized taxi dispatch that jointly enhances earnings and workload equity while preserving operational efficiency.
- It presents, to the best of our knowledge, the first systematic framework for operationalizing multidimensional fairness within reinforcement learning-based taxi dispatch, achieved through Gini-regularized decision-making embedded in action selection, assignment, and reward shaping.
- It conducts an extensive experimental evaluation across two reinforcement learning agents, three demand regimes, and six fairness configurations, providing quantitative evidence on the trade-offs and optimal weighting between earnings and workload fairness under realistic urban mobility conditions.

The remainder of the paper is organized as follows. Section 2 provides a brief review of the key literature on TD, fair TD, and reinforcement learning (RL)-based taxi dispatching. Section 3 outlines the methodological details of the study. Section 4 presents experimental results. Finally, Section 5 concludes the paper by discussing the implications, limitations, and directions for future research.

2. Related-work

2.1.TD Approaches

The TD problem concerns the optimal bipartite matching of available taxis to pending ride requests in a way that maximizes the operational efficiency (Maciejewski et al., 2016; Zhou et al., 2018).

This optimality search frames TD as a combinatorial optimization problem that seeks to find the

best solution from a finite set of combinatorial possibilities involving taxis and ride requests (Zhang et al., 2017). Early approaches often employ rule-based methods for this purpose (Liao, 2003). To exemplify, these methods often involve greedy schemes such as nearest pickup or first-come - first-served, which focus on the local optimal choice in each step. Despite being fast and straightforward, these schemes remain myopic to global solutions and therefore compromise the supply-demand balance in the long run (Miao et al., 2015). Followingly, researchers also employ exact methods to find the best assignment among all possibilities, such as linear programming (Xu & Xu, 2020) and dynamic programming (He, Hu, & Zhang, 2020). One fallback of these methods is that finding the best solution becomes intractable for a large number of taxis and requests at the city scale.

Reinforcement Learning (RL) emerged as a viable approach toward TD with its ability to learn an optimal assignment policy from both experience and historical data and adapt to dynamic and complex environments. Previous research has well established its effectiveness and superiority over its counterparts (Liu et al., 2022; Zhou et al., 2023). There are mainly two RL approaches towards the TD problem: single agent (Liu et al., 2022) and multi-agent (Zhao, Chen, Zhang, & Zhou, 2022). The former is often associated with a centralized approach where a single agent learns the optimal matching between taxis and ride requests in a fully coordinated manner (Hua et al., 2021; Liu et al., 2022). The latter, Multi Agent Reinforcement Learning (MARL), is often associated with a decentralized approach and views taxis as independent agents that learns to make dispatching decisions based on their local observations and shared environment data (Li et al., 2019; Kim & Kim; Rajeh et al., 2024). To alleviate the adversities resulting from an uncoordinated fleet of taxis (Miao et al., 2015), a third strand proposes a hybrid approach (e.g., Huang et al., 2023) that combines centralized approach for global coordination (e.g., demand forecasting,

regional goal setting) with MARL where individual taxis or regional agents learn localized policies, enabling both strategic oversight and adaptive, decentralized execution.

Another key modeling decision in reinforcement learning (RL) concerns the spatial representation of the environment. One common approach partitions the environment into equal-sized grids to represent locations (Nanda et al., 2020; Xie et al., 2023). While this method is easy to encode, particularly with convolutional neural networks, it is susceptible to the Modifiable Areal Unit Problem (MAUP) (Openshaw, 1984), which refers to the variation in analysis results caused by changes in the spatial unit size (De Lima et al., 2020). Moreover, grid-based representations do not reflect the underlying street network, leading to misrepresentations of trip distances and connectivity among locations (Liu et al., 2020). An alternative approach uses graphs to more realistically represent street networks (Tong et al., 2021; Kim & Kim, 2021; Rajeh et al., 2024). Although more complex, graph-based algorithms promise improved agent performance by more accurately capturing the topology of the street network, including realistic distance calculations and connectivity (Davis, Raina, & Jagannathan, 2020).

In sum, an effective solution to the problem of TD should involve the optimal decisions on how to coordinate ride assignments and how to represent the spatial structure of the operation area.

2.2. Fairness-Aware TD

Fairness in TD extends beyond operational efficiency and addresses concerns regarding income inequality among drivers (Sühr et al., 2019; Xu & Xu, 2020; Brar & Su, 2020; Zhao et al., 2022). Unequal earnings can undermine the social welfare of drivers by fostering feelings of envy (Aleksandrov, 2023). When drivers perceive the system as unfair, they are more likely to reject ride requests (Bokányi & Hannák, 2019; Nanda et al., 2020; Guo et al., 2023), which in turn harms

operational efficiency by increasing passenger wait times, reducing customer satisfaction, and ultimately lowering system revenue. To mitigate income inequality, previous studies have proposed various fairness indicators, such as the Generalized Entropy Index (Sühr et al., 2019), standard deviation of earnings (Brar & Su, 2020), and the variance of profit efficiency (earnings per time unit) (Wang et al., 2021). These indicators are incorporated into TD algorithms as part of multi-objective or bipartite optimization functions (Xu & Xu, 2020) that aim to balance system profitability with income fairness.

Another important aspect of fairness in taxi dispatch systems is workload equity, which refers to drivers earning income proportional to their active time on the platform (Sühr et al., 2019). Implementing such a system helps restore drivers' trust and increases their willingness to accept ride requests assigned by the central dispatcher (Bokányi & Hannák, 2019; Nanda et al., 2020; Guo et al., 2023). Equitable systems also reduce the need for drivers to work excessively long hours to meet their earnings goals (Allon et al., 2023). Previous studies have highlighted a strong link between extended working hours and increased risks of fatigue (Lim & Chia, 2015), job stress (Göktaş, 2023), and car accidents (Meng et al., 2015). Therefore, establishing an equitable dispatch system is essential for promoting driver welfare.

In summary, fairness in taxi dispatch systems requires ensuring equal pay for equal workload, thereby satisfying the needs of all stakeholders.

2.3. This study

This study contributes to the fairness-aware taxi dispatch literature by proposing GiniDispatch, a centralized reinforcement learning framework that simultaneously addresses earnings and workload (utilization) equity. While prior research has predominantly focused on income-based

fairness (Sühr et al., 2019; Sun et al., 2022; Liang, 2024) or extended fairness considerations to passenger service quality (Guo et al., 2023), this study treats workload balance as a core driver welfare dimension, closely linked to fatigue, job stress, and operational safety (Allon et al., 2023).

Methodologically, existing reinforcement learning approaches typically operationalize fairness along a single dimension—such as max–min income objectives (Sun et al., 2022) or temporal earnings entropy (Liang, 2024)—whereas GiniDispatch incorporates Gini-regularized equity signals directly into a centralized spatiotemporal dispatch framework, influencing action selection, assignment, and reward shaping in a unified manner. Table 1 summarizes the key research

Table 1. Key research on Fair TD algorithms

Study	Method	Fairness Dimension(s)	Key Fairness Metrics(s)
Sühr et al. (2019)	Two-sided Optimization (ILP)	Driver income, Customer waiting Time	Generalized Entropy Index (GEI)
Nanda et al. (2020)	LP-based Non-adaptive (NAdap)	Driver Matching Fairness	Minimum matching ratio
Xu & Xu (2020)	Bi-objective Linear Program (LP)	Driver Income	
Brar & Su (2020)	Receding Horizon Optimization	Driver Income & Scheduling	Standard deviation of profits; Scheduling Guarantee Ratio (SGR)
Sun et al. (2022)	Multi-Agent Reinforcement Learning (JDRL)	Driver Income	Max-min fairness (maximizing the worst 10% income)
Guo et al. (2023)	Socio-Aware Spatial-Temporal Graph Convolutional Network (SA-STGCN) and Matching-Integrated Vehicle Rebalancing (MIVR)	Algorithmic Fairness, Service Quality Fairness	Mean-Variance of Percentage Error (MVPE), GEI, Standard Deviation of Wait Time
Liang (2024)	RL-based Bipartite Matching (FMRL)	Driver Income, Overall System Income	Weighted amortization fairness entropy
This Study	Gini-regularized STGCRL	Driver Earnings and Workload Equity	Gini Coefficients for Earnings and Utilization (Workload)

3. Methodology

3.1. Problem definition

The problem of Fairness-Aware Taxi Dispatching (FATD) addresses the challenge of dispatching taxis to ride requests while balancing system efficiency with fair treatment of drivers, formulated as a multi-objective optimization problem.

Here, we operationalize fairness using the Gini coefficient, a widely-used measure of inequality ranging from 0 (perfect equality) to 1 (maximum inequality). For a set of values $X = \{x_1, x_2, \dots, x_n\}$, Gini is formulated as follows:

$$G = \frac{\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|}{2n \sum_{i=1}^n x_i} \quad (1)$$

Accordingly, we calculated two separate gini coefficients,

- Earnings fairness measures income distribution equality $G_e = \text{Gini}(\{e_1, e_2, \dots, e_n\})$
- Utilization fairness measures utilization distribution equality $G_u = \text{Gini}(\{u_1, u_2, \dots, u_n\})$

A system is considered fair when both $G_e \rightarrow 0$ and $G_u \rightarrow 0$, indicating that drivers earn proportionally to their working time.

Consider a taxi fleet with n taxis denoted as $T = \{t_1, t_2, \dots, t_n\}$ and m ride requests represented as $R = \{r_1, r_2, \dots, r_m\}$. Each taxi t_i has attributes including current location l_i , accumulated earnings e_i , utilization rate u_i (proportion of time occupied), and availability status a_i . Similarly, each request r_j is characterized by origin location o_j , destination location d_j , fare amount f_j , wait time wt_j , and expected trip duration τ_j . The waiting time, wt_j , for request, r_j , is defined as the elapsed time between the request submission time, t_j^{submit} and taxi pickup time, t_j^{pickup} :

$$wt_j = t_j^{pickup} - t_j^{submit} \quad (2)$$

The traditional dispatch problem seeks to find an optimal assignment matrix $A = [a_{ij}]_{n \times m}$ where $a_{ij} = 1$ if taxi t_i is assigned to request r_j , and 0 otherwise. Each taxi can serve at most one request at a time $\sum_{j=1}^m a_{ij} \leq 1$ for all i , and each request can be assigned to at most one taxi $\sum_{i=1}^n a_{ij} \leq 1$ for all j . The standard objective function maximizes total revenue:

$$Revenue = \sum_{i=1}^n \sum_{j=1}^m a_{ij} \cdot f_j \quad (3)$$

On the other hand, the FATD problem goes beyond the mere revenue (base reward) maximization and involves a multi-objective optimization incorporating fairness. The base reward R_{base} captures the immediate economic value of a taxi-request assignment, computed as:

$$R_{base}(t_i, r_j) = f_j - c_{travel}(l_i, o_j) \quad (4)$$

where f_j is the fare amount and $c_{travel}(l_i, o_j)$ is the travel cost from taxi location l_i to pickup location o_j . Then FATD problem becomes:

$$\max_A \sum_{i,j} a_{ij} \cdot [R_{base}(t_i, r_j) + \gamma \cdot R_{fair}(t_i, r_j)] \quad (5)$$

where $\gamma \geq 0$ is a fairness weight parameter that controls the influence of fairness considerations on the overall objective. It takes on a positive value that scales the importance of fairness relative to the base reward. Higher values of γ amplify the impact of fairness on decision-making. The fairness reward R_{fair} reinforces assignments that reduce system-wide inequality, computed as:

For any dispatch decision, the total reward can be expressed as:

$$R_{fair}(t_i, r_j) = w_e \cdot \Delta G_e + w_u \cdot \Delta G_u \quad (6)$$

Where

- $\Delta G_e = G_e^{before} - G_e^{after}$ is the earnings Gini coefficient reduction
- $\Delta G_u = G_u^{before} - G_u^{after}$ is the utilization Gini coefficient reduction
- w_e, w_u are earnings and utilization weights respectively.

The fairness reward is dynamically based on current inequality levels:

$$R_{fair}^{scaled} = R_{fair} \cdot e^{\alpha \cdot G_{current}} \quad (7)$$

Where α denotes the correction intensity, applying stronger fairness when inequality is high.

3.2. Building the Street Graph

We construct a dynamic bipartite graph $G_t = (V, E)$ at each timestep, where $V = T \cup R$ comprises taxi nodes T and request nodes R . The edge set E forms a fully connected bipartite structure with bidirectional edges between all taxi-request pairs, enabling message passing in both directions. No distance-based thresholding or sparsification is applied; edge weights are uniform ($w_{ij} = 1$).

Taxi nodes encode a 4-dimensional feature vector: $[locationzone, availability, cumulative\ earnings, busy\ time]$. Request nodes encode: $[pickupzone, dropoffzone, fare, duration]$. Temporal features $[hour/24, minute/60, weekday/6]$ are appended to all nodes.

We pre-compute a distance matrix $D \in \mathbb{R}^{|Z| \times |Z|}$ from historical trip records, where $D_{o,d} = \text{mean}(\text{trip_distance})$ for all trips between zones o and d . For missing pairs, we apply cascading fallbacks: reverse lookup ($\times 1.1$ penalty), zone-averaged distances, or ID-based scaling. Travel time uses actual trip duration from the dataset; taxi repositioning assumes 3 min/mile.

The STGCRL comprises two GCNConv layers (128 hidden units each) with batch normalization, ReLU activation, and residual connections. Node embeddings are aggregated via global mean pooling, concatenated with taxi and encoded request features, then passed through an MLP to predict Q-values for action selection.

3.3. The GiniDispatch Algorithm

The GiniDispatch algorithm operates in three coordinated steps.

Step 1: Action Selection with Position-Based Fairness Boosts

The first step determines which ride request $r_i \in R = \{r_1, r_2, \dots, r_m\}$ should be prioritized from the current pool of active requests. The process begins by training Q-values from the trained Q-network given the current state s . These Q-values represent the expected cumulative reward for selecting each request. The algorithm extracts valid Q-values for the available actions. When the fairness weight $w_f > 0$, the algorithm computes position-based fairness boosts for each request. It first calculates fleet-wide statistics: mean earnings, $\bar{e}$, and mean utilization, $\bar{u}$, and their respective ranges, range_e and range_u . The current Gini coefficients, G_e^{current} and G_u^{current} are computed, and dynamic scaling factors are derived as α_e and α_u . For each request, r_i , the algorithm identifies eligible taxis and evaluates the best fairness impact. For each eligible taxi j , the position scores are computed, pos_e^j and pos_u^j . The position-based boosts (i.e., boost_e^j boost_u^j) are the negatives of these positions, meaning taxis below the fleet average receive positive boosts. A distance penalty, $\text{penalty}_{\text{dist}}^j$, is applied, and the combined fairness is calculated. The final action selection combines Q-values with fairness adjustments. The details regarding the calculations is displayed in Table 2 below.

Table 2. Step 1:-Action Selection with Position-Based Fairness Boosts - Pseudocode

Step 1 The GiniDispatch Algorithm: Action Selection	
	Require: Current state s , Q-network, active requests $R = \{r_1, \dots, r_m\}$
	Require: Taxi fleet T with earnings E , utilization U , fairness weight w_f
	Step 1: Get Q-Values
1:	$q_values \leftarrow \text{Q-Network}(s)$
2:	$valid_actions \leftarrow \min(R , action_size)$
3:	$valid_q_values \leftarrow q_values[0 : valid_actions]$
4:	if $w_f > 0$ then
	Step 2: Calculate Position-Based Fairness Impacts
5:	$\bar{e} \leftarrow (1/ T) \sum_{t \in T} e_t, range_e \leftarrow \max(E) - \min(E)$
6:	$\bar{u} \leftarrow (1/ T) \sum_{t \in T} u_t, range_u \leftarrow \max(U) - \min(U)$
7:	$G_e^{current} \leftarrow \text{ComputeGini}(E), \alpha_e \leftarrow \exp(\min(G_e^{current} \times 3, 3))$
8:	$G_u^{current} \leftarrow \text{ComputeGini}(U), \alpha_u \leftarrow \exp(\min(G_u^{current} \times 3, 3))$
9:	$fairness_boosts \leftarrow \text{zeros}(valid_actions)$
10:	for $i = 0$ to $valid_actions - 1$ do
11:	$r_i \leftarrow R[i]$ {Get request i }
12:	$eligible_taxis \leftarrow \text{FindEligibleTaxis}(r_i.pickup_loc)$
13:	$best_fairness_impact \leftarrow -\infty$
14:	for each $(taxi_j, dist_j) \in eligible_taxis$ do
15:	$pos_e^j \leftarrow (e_j - \bar{e}) / range_e$ {Earnings position}
16:	$pos_u^j \leftarrow (u_j - \bar{u}) / range_u$ {Utilization position}
17:	$boost_e^j \leftarrow -pos_e^j$ {Position-based earnings boost}
18:	$boost_u^j \leftarrow -pos_u^j$ {Position-based utilization boost}
19:	$penalty_{dist}^j \leftarrow 1.0 / (1.0 + dist_j \times 0.1)$ {Distance penalty}
20:	$fairness_impact_e^j \leftarrow boost_e^j \times \alpha_e \times penalty_{dist}^j$
21:	$fairness_impact_u^j \leftarrow boost_u^j \times \alpha_u \times penalty_{dist}^j$
22:	$fairness_impact^j \leftarrow fairness_impact_e^j + fairness_impact_u^j$
23:	$best_fairness_impact \leftarrow \max(best_fairness_impact, fairness_impact^j)$
24:	end for
25:	$fairness_boosts[i] \leftarrow best_fairness_impact \times fairness_scale \times w_f$
26:	end for
	Step 3: Select Action with Fairness Adjustment
27:	$adjusted_q_values \leftarrow valid_q_values + fairness_boosts$
28:	$action \leftarrow \arg \max(adjusted_q_values)$
29:	else
	Step 3: Select Action without Fairness
30:	$action \leftarrow \arg \max(valid_q_values)$
31:	end if
32:	return $action$

Step 2: Taxi Assignment

Once a request r is selected, GiniDispatch assigns the optimal taxi from the eligible set. The set includes all taxis from the fleet T that are currently available (not serving another passenger) and located within a distance threshold θ from the request's pickup location, loc_p . When fairness is enabled ($w_f > 0$), the algorithm performs Gini simulation for each eligible taxi t_j . For earnings impact, it creates a simulated earnings vector, E^{new} by adding the fare f and then computes the impact by taking the difference between $fairness_E^{after}$ and $fairness_E^{before}$. Similarly, for utilization impact with trip duration d , it computes by taking the difference between $fairness_U^{after}$ and $fairness_U^{before}$. The weighted assignment score, $score_j$, combines both impacts with distance penalty, $penalty_{dist}^i$. The taxi with the highest score is selected. When fairness is disabled, the algorithm defaults to nearest taxi assignment using $score_j$, negative distance value, $-dist_j$. The details regarding the calculations is displayed in Table 3 below.

Table 3. Step 2:Taxi Assignment - Pseudocode

Step 2 The GiniDispatch Algorithm: Taxi Assignment	
	Require: Selected request r with pickup loc_p , fare f , duration d Require: Taxi fleet T with current earnings E , utilization U Require: Fairness weights w_e , w_u , fairness weight w_f Step 1: Find Eligible Taxis 1: $T_{eligible} \leftarrow \{t \in T : t \text{ available AND Distance}(t.loc, loc_p) \leq threshold\}$ 2: if $w_f > 0$ AND fairness enabled then Step 2: Gini Simulation for Each Eligible Taxi 3: $taxi_scores \leftarrow \emptyset$ 4: for each taxi $t_j \in T_{eligible}$ do 2a: Simulate Earnings Impact 5: $E^{new} \leftarrow E; E^{new}[j] \leftarrow E^{new}[j] + f$ 6: $fairness_e^{before} \leftarrow 1 - \text{ComputeGini}(E)$ 7: $fairness_e^{after} \leftarrow 1 - \text{ComputeGini}(E^{new})$ 8: $impact_e^j \leftarrow fairness_e^{after} - fairness_e^{before}$ 2b: Simulate Utilization Impact 9: $U^{new} \leftarrow U; U^{new}[j] \leftarrow U^{new}[j] + d / total_time$ 10: $fairness_u^{before} \leftarrow 1 - \text{ComputeGini}(U)$ 11: $fairness_u^{after} \leftarrow 1 - \text{ComputeGini}(U^{new})$ 12: $impact_u^j \leftarrow fairness_u^{after} - fairness_u^{before}$ 2c: Calculate Weighted Score 13: $dist_j \leftarrow \text{Distance}(t_j.loc, loc_p)$ 14: $penalty_{dist}^j \leftarrow 1.0 / (1.0 + dist_j \times 0.1)$ 15: $score_j \leftarrow (w_e \times impact_e^j + w_u \times impact_u^j) \times penalty_{dist}^j$ 16: $taxi_scores \leftarrow taxi_scores \cup \{(t_j, score_j)\}$ 17: end for 18: else Step 2: Distance-Only Assignment 19: for each taxi $t_j \in T_{eligible}$ do 20: $dist_j \leftarrow \text{Distance}(t_j.loc, loc_p)$ 21: $taxi_scores \leftarrow taxi_scores \cup \{(t_j, -dist_j)\}$ 22: end for 23: end if Step 3: Select Best Taxi 24: $t_{selected} \leftarrow \arg \max(t_j, score_j) \in taxi_scores$ 25: return $t_{selected}$

Step 3: Fairness Reward Calculation

This calculates the reward signal used to train the DQN by combining service efficiency with fairness considerations. The algorithm first computes a base reward based on the trip fare, f , adjusting it with bonuses for timely service and penalties for excessive passenger wait times. When fairness is enabled ($w_f > 0$), the algorithm simulates how assigning the trip to the selected taxi affects the overall earning, $E^{\text{after}}[\text{sel}]$, and utilization, $U^{\text{after}}[\text{sel}]$ across the fleet. It measures the change in fairness (i.e., $1 - G(\cdot)$) before and after the assignment to quantify the fairness impacts, impact_e and impact_u . The algorithm then applies dynamic scaling factors, scale_e and scale_u , that amplify the importance of whichever dimension currently exhibits greater inequality. These fairness impacts are combined using dynamically normalized weights $\text{dynamic_}w_e$ and $\text{dynamic_}w_u$ into a fairness reward component, which is scaled by the current inequality severity and added to the base reward. Finally, the total reward is clipped to the range $[-10, 100]$ to ensure the stable training of the neural networks. The details regarding the calculations is displayed in Table 4 below.

3.4. RL Agents

We employ two distinct RL agents to evaluate the generalizability of GiniDispatch across different learning paradigms. STGCRL represents state-of-the-art deep RL with graph neural networks, capable of capturing spatial dependencies and road network topology (Kim & Kim, 2021). Its continuous state representation enables nuanced learning of complex dispatch patterns. Q-Learning serves as a traditional tabular baseline with discretized states, testing whether fairness improvements persist in simpler learning frameworks without deep function approximation.

Table 4. Step 3: Fairness Reward Calculation - Pseudocode

Step 3 The GiniDispatch Algorithm: Fairness Reward Calculation	
	Require: Assigned taxi t_{sel}, request r with fare f, duration d Require: Current system state: earnings E, utilization U before assignment Require: Fairness weights w_e, w_u, fairness weight w_f Step 1: Calculate Base Reward 1: $wait_time \leftarrow current_time - r.system_receive_time$ 2: $priority_bonus \leftarrow \min(r.priority / 5, 1.0) \times on_time_bonus$ 3: if $wait_time > wait_threshold$ then 4: $penalty \leftarrow wait_penalty_coef \times (wait_time - wait_threshold)$ 5: $base_reward \leftarrow f - penalty + priority_bonus$ 6: else 7: $base_reward \leftarrow f + on_time_bonus + priority_bonus$ 8: end if 9: if $w_f > 0$ AND fairness enabled then Step 2: Calculate Fairness Reward using Gini Simulation 2a: Record Current Fairness State 10: $fairness_e^{before} \leftarrow 1 - \text{ComputeGini}(E)$ 11: $fairness_u^{before} \leftarrow 1 - \text{ComputeGini}(U)$ 12: $gini_e^{before} \leftarrow 1 - fairness_e^{before}$, $gini_u^{before} \leftarrow 1 - fairness_u^{before}$ 2b: Update System State 13: $E^{after}[sel] \leftarrow E^{after}[sel] + f$ 14: $U^{after}[sel] \leftarrow U^{after}[sel] + d / total_time$ 2c: Calculate New Fairness State 15: $fairness_e^{after} \leftarrow 1 - \text{ComputeGini}(E^{after})$ 16: $fairness_u^{after} \leftarrow 1 - \text{ComputeGini}(U^{after})$ 2d: Calculate Fairness Impact 17: $impact_e \leftarrow fairness_e^{after} - fairness_e^{before}$ 18: $impact_u \leftarrow fairness_u^{after} - fairness_u^{before}$ 2e: Apply Dynamic Scaling 19: $scale_e \leftarrow \exp(\min(gini_e^{before} \times 3, 3))$ 20: $scale_u \leftarrow \exp(\min(gini_u^{before} \times 3, 3))$ 21: $total_weight \leftarrow w_e \times scale_e + w_u \times scale_u$ 22: $dynamic_w_e \leftarrow (w_e \times scale_e) / total_weight$, $dynamic_w_u \leftarrow (w_u \times scale_u) / total_weight$ 2f: Calculate Fairness Reward 23: $fairness_component \leftarrow dynamic_w_e \times impact_e + dynamic_w_u \times impact_u$ 24: $fairness_severity \leftarrow \max(gini_e^{before}, gini_u^{before})$ 25: $fairness_scale \leftarrow 20.0 \times \min(1.0 + fairness_severity, 3.0)$ 26: $fairness_reward \leftarrow fairness_component \times fairness_scale \times w_f$ Step 3: Calculate Final Reward 27: $total_reward \leftarrow base_reward + fairness_reward$ 28: else 29: $total_reward \leftarrow base_reward$

30:	end if
	Step 4: Clip Reward to Reasonable Range
31:	$total_reward \leftarrow \max(-10.0, \min(total_reward, 100.0))$
32:	return $total_reward$

This comparison addresses whether fairness-aware dispatch requires sophisticated spatial modeling or can be achieved with classical methods, informing practical deployment decisions for platforms with varying computational resources.

Spatiotemporal Graph Convolutional Reinforcement Learning (STGCRL) Agent

Before developing our STGCRL agent, we constructed a data-driven street graph from the NYC Taxi dataset. Each node represents a unique location ID, with edges weighted by average distances between locations. We extracted unique pickup and dropoff location IDs, grouped ride requests by location pairs, and calculated average trip distances as edge weights in the adjacency matrix. This resulted in a matrix with 262 nodes and 25497 edges, and an average weight (internodal distance) of 11.06 miles.

Our STGCRL agent models the taxi environment as a dynamic graph $G_t = (V_t, E_t)$ at timestep t , where node set V_t comprises taxi nodes V_{taxi} and request nodes V_{request} , with edge set E_t connecting potential matches. The agent processes spatiotemporal features through three stacked graph convolutional layers that capture both local taxi-request interactions and global system patterns through neighborhood aggregation $N_{(i)}$.

Temporal dynamics are incorporated by constructing feature vectors $x_i = [s_i; t_i]$ for each node $v_i \in V_t$, concatenating spatial coordinates s_i with temporal features t_i including time-of-day and waiting

Table 5. Key STGCRL Parameters

Key Parameters	Description	Value
<i>Time step duration</i>	Time increment per simulation step (minutes)	1
<i>Learning rate</i>	Gradient descent step size for neural network updates (α)	0.03
<i>Max Requests considered</i>	Maximum number of active requests processed per time step	200
<i>Discount factor</i>	Future reward decay rate (γ , typically 0.9-0.99)	0.99
<i>Batch size</i>	Number of experiences sampled for each training update	32
<i>Epsilon</i>	Exploration probability in ϵ -greedy action selection	1
<i>Epsilon decay</i>	Rate of exploration reduction over training episodes	0.992
<i>Epsilon min</i>		0.05
<i>Patience</i>	Episodes to wait without improvement before early stopping	10
<i>Minimum delta</i>	Minimum reward improvement threshold for early stopping	0.01
<i>Wait threshold</i>	Maximum passenger wait time before penalty applies (minutes)	5
<i>On-time bonus</i>	Reward bonus for completing requests within wait threshold	5
<i>Number of episodes</i>	Total training/evaluation episodes per configuration	100/20
<i>Max_wait</i>	The maximum waiting time before a request expires	30

duration. After graph convolution, global mean pooling aggregates node features into a graph-level representation, which is then fused with entity-specific taxi and request characteristics.

The STGCRL framework defines the state space S as graph representations G_t , action space A as taxi-request assignments, and reward function $r(s_t, a_t, s_{t+1})$ based on assignment utility. The agent approximates the Q -function $Q(s_t, a_t)$ for potential assignments, leveraging spatiotemporal representations to consider both the current system state and temporal evolution. This integration of graph neural networks with reinforcement learning exploits structural relationships between entities and the sequential nature of the assignment process for effective decision-making in dynamic environments. Table 5 displays the key parameters of the STGCRL agent. Notably, the changing parameter values across request sizes were determined based on the experiments with several values for each sample size.

Q-Learning Agent

Our fairness-aware Q-learning agent uses a traditional tabular approach for ride assignment decisions. The agent maintains a Q-table that maps discretized state representations to action values for request selection decisions. The state representation uses location binning to manage complexity, where taxi and request locations are discretized into bins. Each state combines available taxi locations (sorted tuple of the first 3 available taxis) with the highest priority active request information, creating state keys that pair available taxi states with request states. Action space A comprises indices of active ride requests, where each action corresponds to selecting which request from the active request list to process. The agent uses ϵ -greedy exploration with optional fairness-aware action selection that applies fairness boosts to Q-values when fairness information is provided.

The fairness-aware Q-learning framework incorporates fairness through action selection modification rather than learning updates. When fairness is enabled, the agent receives fairness impact information from the environment and scales these impacts to adjust Q-values before action selection, prioritizing assignments that benefit underperforming taxis. The agent learns through standard Q-learning temporal difference updates, while the environment independently handles request assignment mechanics, fairness calculations, and reward computation. Table 6. lists key parameters unique to Q Learning agent. The other parameters remain unchanged across the agents.

Table 6. Key Parameters Unique to Q-Learning Agent

Key Parameters	Description	Value
<i>State discretization method</i>	How continuous states are binned for Q-table	Location binning
<i>State key components</i>	Elements used to create state representation	(taxi_locations, request_info)
<i>Q-table initialization</i>	How Q-values are initially set	Zeros
<i>Action space limitation</i>	How actions are constrained to valid requests	min(active_requests, action_size)
<i>TD error tracking</i>	Whether temporal difference errors are monitored	Yes

Baselines

We implemented two baseline dispatching strategies to establish performance benchmarks against our Drive Equalizer algorithm: the Greedy-Nearest baseline assigns each incoming ride request to the nearest available taxi based on pickup distance, representing a simple reactive dispatching heuristic commonly used in practice, while the Hungarian-Optimal baseline formulates the assignment problem as a bipartite matching task and employs the Hungarian algorithm to find the globally optimal assignment that minimizes total pickup distances across all active requests and available taxis in each time step. Both baselines operate with a discrete time step of 1 minute and implement a wait penalty mechanism where requests waiting longer than 5 minutes incur a penalty of 0.5 per additional minute, while requests dispatched within the threshold receive a bonus of 5 reward units, with requests exceeding 30 minutes of wait time being rejected from the system. We evaluated both baselines across three demand levels (250, 750, and 1500 requests) with a fleet of 250 taxis, conducting 20 episodes for both training and evaluation phases with a 50/50 train-eval split, introducing temporal noise (± 2 minutes) to request timestamps during training episodes to simulate realistic operational variability. Performance metrics included episode reward (fare with

wait penalties/bonuses), base reward (fare only), earnings fairness (Gini coefficient across taxi earnings), utilization fairness (Gini coefficient across taxi busy time), completed trips, average wait time, rejected requests, and trip efficiency (multiplication of the number of completed trips in an episode by the inverse of the utilization rate).

3.5. Real-Time Simulation Setup

The environment simulates a real-world taxi dispatching scenario using a sample from the NYC taxi dataset that spans all taxi transactions during January 2024 ($n=2964624$). To ensure valid values across trip duration and distance, we performed an outlier analysis and removed 453271 transactions, resulting in a total of 2511353 transactions. After obtaining a clean dataset, we chronologically ordered the requests by pickup time used a sample from the January 1st, early hours of the new year.

The fleet size of 250 taxis was selected based on three considerations. First, the NYC Taxi dataset sampling window (early hours of January 1st) exhibits moderate demand density, with approximately 100 requests per minute at peak times. A fleet of 250 taxis at this demand level produces meaningful dispatch competition without trivial oversupply or severe undersupply—enabling observation of fairness dynamics under realistic operational stress. Second, this configuration aligns with prior simulation studies in taxi dispatch research: Wang et al. (2021) used 200 taxis, Nanda et al. (2020) employed 100-500 taxi ranges, and Rajeh et al. (2024) tested with 300 vehicles. Third, the Requests-Per-Taxi (RPT) ratios of 1 (low), 3 (moderate), and 6 (high)

at 250 taxis create meaningful variation in dispatch pressure while remaining computationally tractable for the 36 experimental configurations ($2 \text{ agents} \times 3 \text{ demand levels} \times 6 \text{ fairness settings}$).

Simulation Environment

Each episode simulates real-time dispatch over approximately 1 hour. At each timestep (1 minute), the environment: (1) generates new requests based on historical arrival patterns, (2) updates taxi availability based on trip completion, (3) processes batch assignments for pending requests, and (4) removes expired requests.

Temporal Split

To ensure generalization, we use temporal train-evaluation splits. Training uses the first 50% of chronologically-ordered requests; evaluation uses the subsequent 50%, starting after the latest training dropoff time to ensure all taxis are available.

Next, we evaluated the performance trained agents on the same demand levels with requests that are placed after the latest dropoff time in the train requests. This is done to ensure the availability of the whole taxi fleet in the beginning of evaluation phase where we put the agents to work in

exploitation mode by setting the ϵ to 0. Figure 1 shows the demand distributions across different demand levels marked by the latest dropoff time.

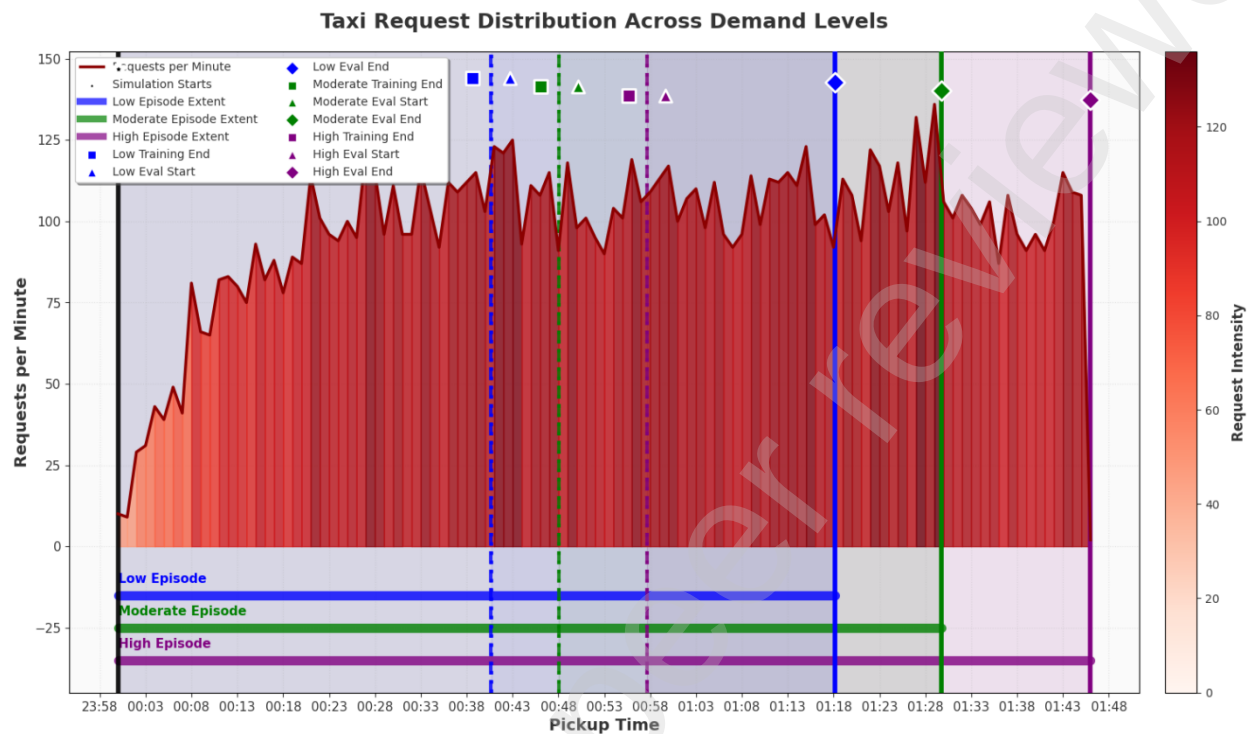


Figure 1. Taxi Demand Distribution Across Episodes

The simulation experiments test various weight configurations (Table 7) for three key parameters: earnings weight (which prioritizes fairness in earnings distribution), utilization weight (which emphasizes equal work distribution), and fairness weight (which determines the overall importance of fairness in the reward function). Notably, while the earnings and utilization weights sum to 1, after several experiments, we decided to fix the value of the fairness weight (w_f) at 5.

Table 7. The fairness configurations

Configuration Name	Earnings Weight (w_e)	Utilization Weight (w_u)	Fairness Weight (w_f)
<i>No Fairness (NF)</i>	0	0	0
<i>Earnings Only (EO)</i>	1	0	5
<i>Strong Earnings Bias (SEB)</i>	0.8	0.2	5
<i>Balanced (B)</i>	0.5	0.5	5
<i>Strong Utilization Bias (SUB)</i>	0.2	0.8	5
<i>Utilization Only (U0)</i>	0	1	5

The performance of the RL agents across different fairness weight configurations and demand levels was evaluated based on operational efficiency (i.e., total system reward, base reward, and waiting time) and fairness (i.e., Gini coefficients of earnings and utilization). Gini coefficients are computed at episode end using cumulative earnings and utilization rates. Utilization rate (eq. 8) tracks the proportion of time taxi t_i spent serving passengers.

$$u_i = \frac{busy_time_i}{episode\ duration} \quad (8)$$

Additionally, we introduce a supplementary metric, Trip Efficiency that is calculated by multiplying the number of completed trips in an episode by the inverse of the utilization rate, highlighting the agents' ability to complete trips with efficient utilization.

To quantify the practical significance of metric performance differences across fairness configurations beyond percentage-based comparisons, we employ Cohen's d as the primary effect

size measure (Cohen, 2016). Cohen's d is widely adopted in experimental research for comparing group means on a standardized scale, expressing differences in units of pooled standard deviation and enabling direct comparison across metrics with heterogeneous scales (Lakens, 2013). Because raw performance values span vastly different ranges across demand levels—base reward ranges from approximately 5,000 to 15,000 while Gini coefficients operate between 0 and 0.4—pooling observations directly would inflate the denominator and mask meaningful configuration-driven variation. We therefore normalize values within each algorithm $\times$ phase $\times$ demand level combination prior to computing pairwise effect sizes, isolating the contribution of fairness configurations from scale-induced variance (Cumming, 2013). Effect sizes are interpreted following conventional thresholds: negligible ($|d| < 0.2$), small ($0.2 \leq |d| < 0.5$), medium ($0.5 \leq |d| < 0.8$), and large ($|d| \geq 0.8$) (Cohen, 2016). This approach complements the percentage improvement analysis by providing a standardized measure of separation between fairness configurations that is invariant to the absolute scale of each metric, allowing us to assess whether observed improvements constitute practically meaningful effects or fall within the range of natural variation across experimental conditions (Sullivan & Feinn, 2012).

4. The Experiment Results

The first set of results demonstrates the efficacy of the proposed agents, STGCRL and Q-learning. By calculating the improvements in total reward between the training and evaluation phases, we validate the performance of these agents, as evidenced by the positive improvements displayed in Figure 2. Both agents consistently increased total rewards across all fairness configurations and demand volumes. On average, the STGCRL and Q-learning agents achieved total reward improvements of 5.48% and 5.73%, respectively. These gains provide strong evidence of the

effectiveness of reinforcement learning–based agents (Gros et al., 2023) and support the validity of the subsequent results.

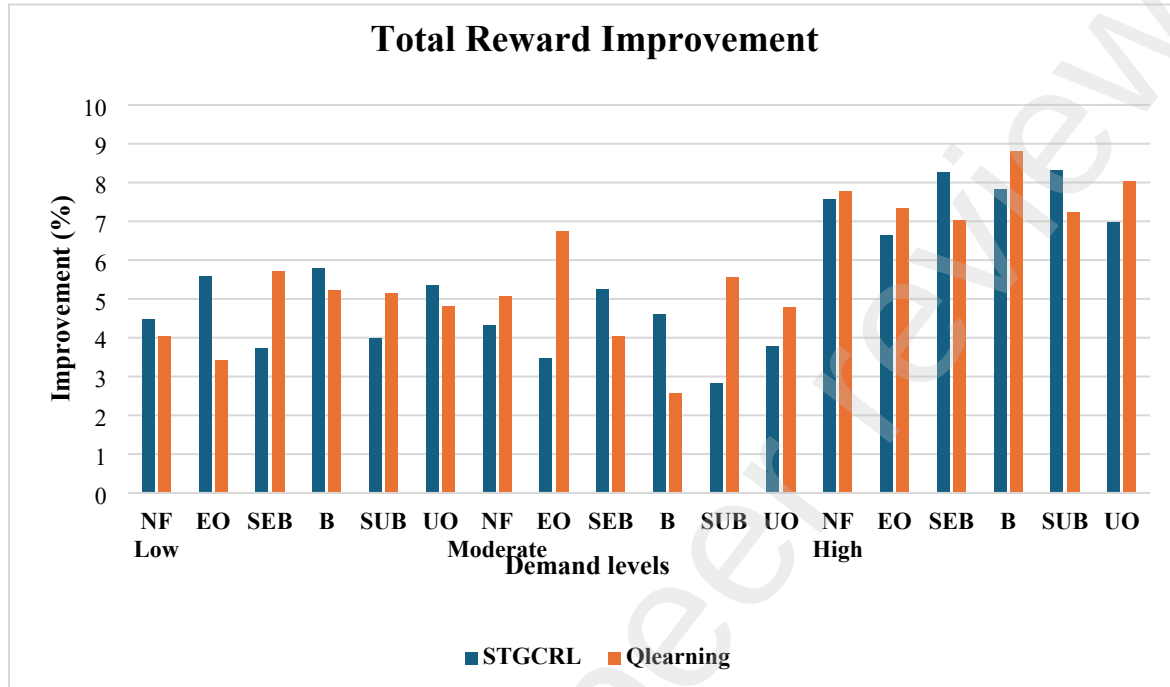


Figure 2. Total reward improvement between training and evaluation phases

Figure 3 compares the improvements achieved by the agents across various configurations relative to the NF (No Fairness) configuration. By removing the fairness reward, we were able to directly compare the reward performance of fairness-aware agents with that of non-fair agents. The results suggest that the TD system does not need to sacrifice profitability in pursuit of fairness. In some configurations, the agents even achieved reward improvements of up to 2% as evidenced in (Nanda et al., 2020; Xu & Xu, 2020).

Figure 3 presents spider charts comparing the mean performance of the proposed RL agents (STGCRL and Q-Learning) against two non-learning baselines (i.e., Nearest Greedy and Hungarian Algorithm) across fairness configurations, where all metrics are min–max normalized

and axes for earnings Gini, utilization Gini, and waiting time are inverted so that higher values consistently indicate better performance. In both training and evaluation phases, the RL agents produce visibly larger and more balanced polygons, dominating the baselines across nearly all dimensions. The disparity is most pronounced along the reward and base reward axes, where both STGCRL and Q-Learning consistently reach the upper bounds of the normalized scale while the baselines remain clustered near or below the midpoint — directly confirming that the learned dispatch policies accumulate substantially higher rewards than rule-based assignment heuristics. Crucially, this reward superiority does not come at the expense of equity: the RL agents simultaneously achieve markedly lower earnings and utilization Gini coefficients, yielding broader coverage on the fairness axes where the baselines collapse. The only dimension on which the Nearest Greedy strategy remains competitive is waiting time, an expected artifact of its myopic distance-minimizing logic (Liang, 2024), yet this narrow advantage is vastly offset by its poor reward and fairness outcomes (Nanda et al., 2020). The consistent replication of these patterns from training to evaluation further confirms the generalization capacity of the proposed agents, collectively establishing that fairness-aware reinforcement learning provides a comprehensively superior dispatch strategy over conventional non-learning assignment methods (Li et. al., 2019; Guo et al., 2023).

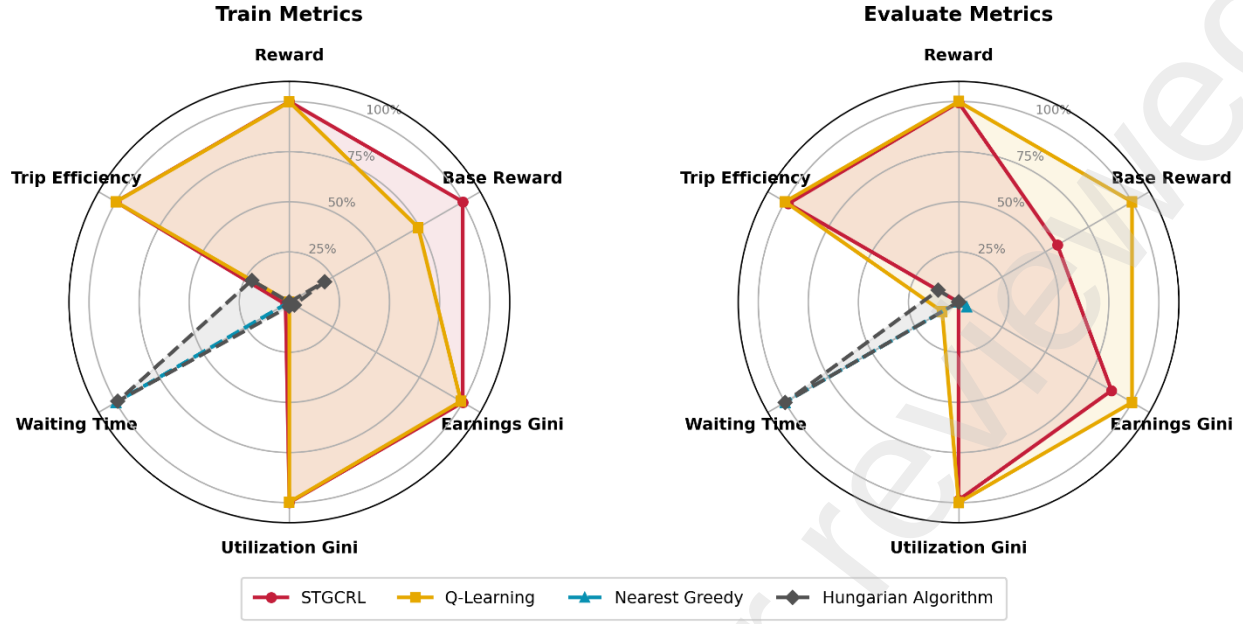


Figure 3. Spider charts comparing RL agents with nonlearning baselines

Figure 4 illustrates the relative base reward improvements achieved by fairness-aware agents across fairness configurations and demand volumes. Normalized Cohen's d effect sizes between the no-fairness baseline (NF) and each fairness configuration confirm that fairness-aware assignment does not systematically degrade base rewards. For STGCRL, all pairwise effect sizes remain small to negligible ($|d| \leq 0.43$), indicating that fairness constraints introduce minimal reward disruption—consistent with the agent's capacity to exploit spatiotemporal correlations that buffer dispatch quality (Wu et al., 2019; Yu & Hu, 2021). Q-learning, by contrast, shows greater sensitivity to the choice of fairness mechanism, with UO producing a large effect favoring the fairness configuration ($d = -1.19$) and SEB, B, and SUB all reaching medium effect sizes ($d = -0.61$ to -0.68), suggesting that fairness constraints actively redirect Q-learning toward higher-reward policies rather than degrading performance. Importantly, the negative direction of these effects across both agents indicates reward improvements rather than losses, and no STGCRL comparison exceeds the conventional small-effect threshold, reinforcing that fairness can be

incorporated into reinforcement learning frameworks without systematically sacrificing efficiency (Raman et al., 2021).

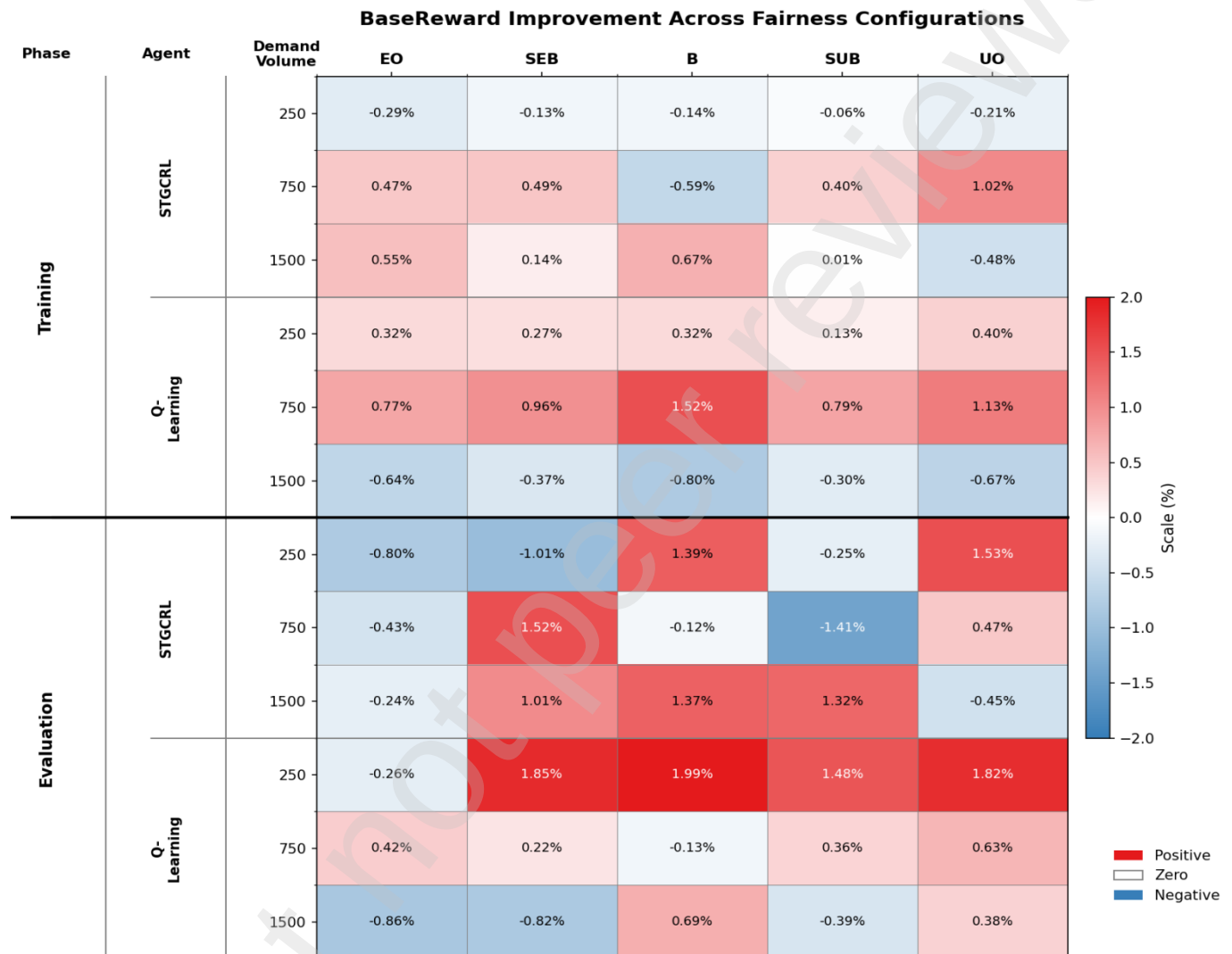


Figure 4. Base reward improvements achieved by fair agents across demand volumes

Passenger waiting time shows mixed responses to fairness interventions, and normalized Cohen's d effect sizes confirm this heterogeneity. During training, Q-Learning at 250 requests achieves substantial waiting time reductions (3–7% improvements across configurations), while STGCRL shows smaller gains; at 1500 training requests, both agents achieve modest improvements (0.8–1.2%). The largest effects emerge for Q-Learning under EO ($d = 0.61$, medium), indicating that

earnings-focused fairness meaningfully reduces passenger waiting times, while STGCRL exhibits a medium effect under SUB ($d = -0.70$) in the opposite direction, suggesting that sequential utilization balancing can increase wait times as drivers are redistributed away from demand-dense areas. The remaining configurations produce predominantly small effects ($|d| = 0.24\text{--}0.49$), with no clear directional consensus—STGCRL-SEB and STGCRL-UO show small positive effects ($d \approx 0.45\text{--}0.49$) favoring reduced waiting times, while STGCRL-B and STGCRL-EO yield small negative effects. Evaluation reveals near-zero impact at 250 requests for both agents, with mixed results at higher demand levels. STGCRL-UO achieves 2.51% improvement at 1500 evaluation requests, suggesting that utilization-focused fairness can enhance service quality under high-demand conditions by promoting balanced driver spatial distribution (Sühr et al., 2019; Guo et al., 2023).

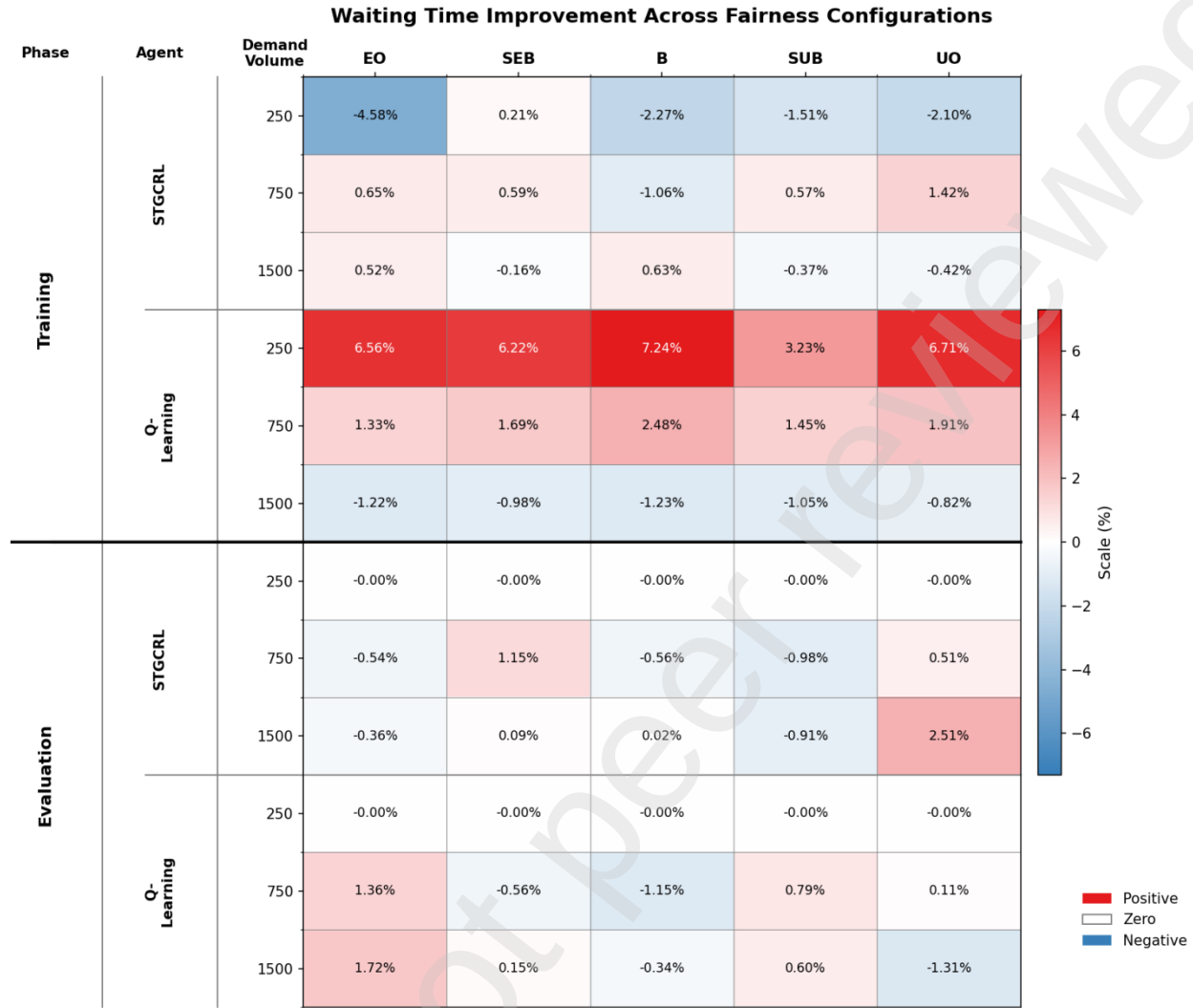


Figure 5. Waiting time improvements achieved by fair agents across demand volumes

Trip efficiency results (see Figure 6) demonstrate consistently positive responses to fairness configurations across evaluation conditions, with improvements ranging from 0.5–2.5% over the no-fairness baseline. Normalized Cohen's d analysis reveals uniformly large effect sizes across all fairness comparisons, with every NF-versus-fairness pair exceeding the large-effect threshold ($|d| > 0.8$) for both agents. STGCRL exhibits the strongest effects under B ($d = -3.32$) and SEB ($d = -2.53$), where the Balanced configuration achieves 2.50% improvement at 250 evaluation requests, while Q-Learning shows its largest effects under SUB ($d = -1.99$) and UO ($d = -1.79$).

The consistently negative direction of all effect sizes confirms that fairness constraints systematically improve rather than compromise efficiency—a counterintuitive finding suggesting that equitable driver distribution across the service area reduces deadheading distances and improves pickup-to-destination ratios (Shi et al., 2021; Sun et al., 2021). Notably, STGCRL produces larger effect magnitudes than Q-Learning across all configurations ($|d| = 1.21\text{--}3.32$ versus $1.07\text{--}1.99$), indicating that spatiotemporal graph representations amplify the efficiency benefits of fairness-aware dispatch. The stronger improvements at lower demand volumes indicate that fairness mechanisms provide the greatest efficiency benefits when task scarcity would otherwise lead to suboptimal spatial clustering of active drivers (Iglesias et al., 2018).

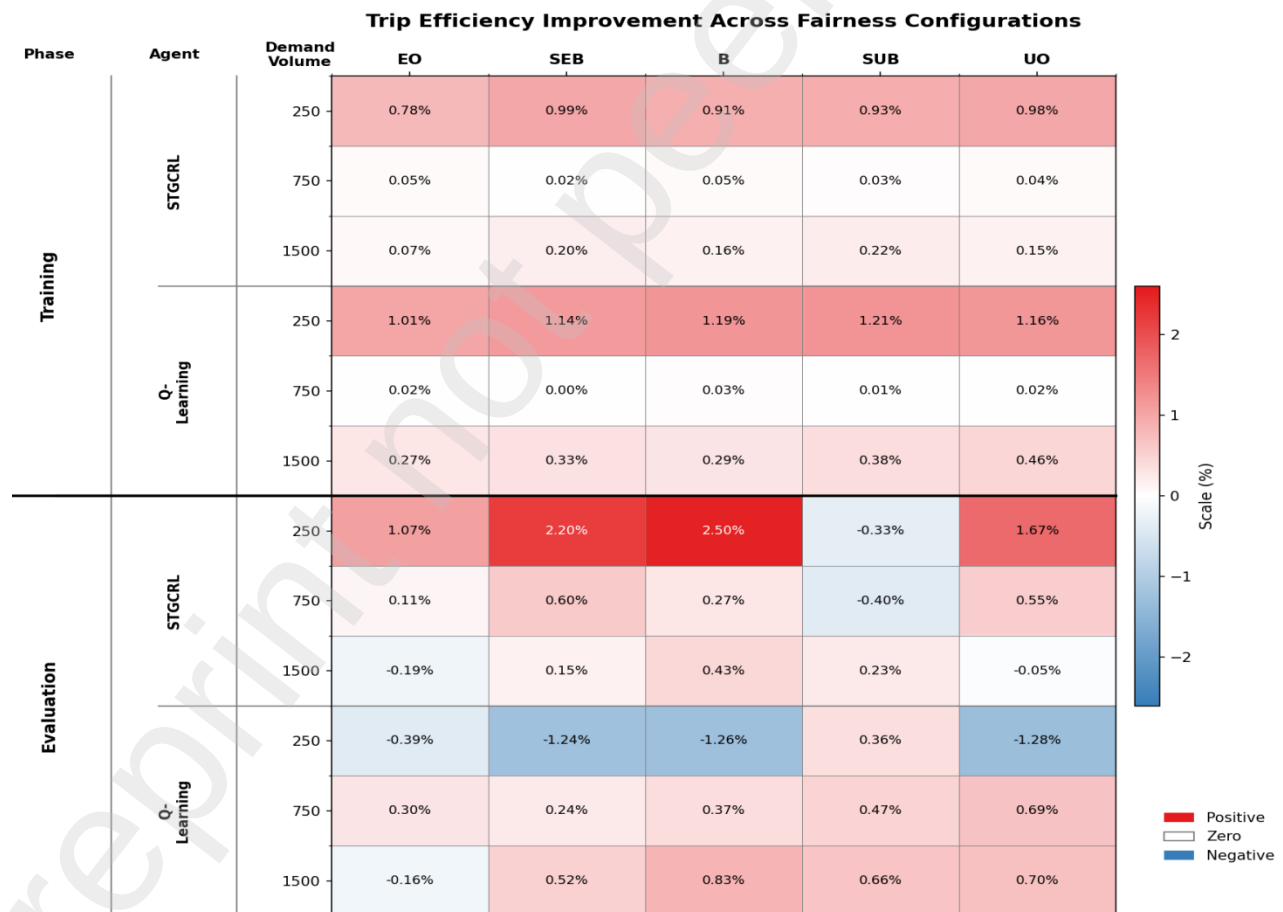


Figure 6. Trip efficiency improvements achieved by fair agents across demand volumes

Figure 7 shows that all fairness configurations achieve substantial earnings inequality reductions during training, with EO delivering the strongest improvements at 750 requests (STGCRL: 34.26%, Q-Learning: 33.79%). Normalized Cohen's d analysis confirms that these reductions are practically significant, with STGCRL producing large effects across all fairness comparisons—SEB ($d = 3.99$) and UO ($d = 3.90$) representing the strongest separations from the no-fairness baseline, followed by B ($d = 3.75$) and SUB ($d = 2.69$). The consistently positive direction indicates that NF maintains higher Gini values (greater inequality) than every fairness configuration, validating the intended mechanism. These training gains generalize well for STGCRL in evaluation, maintaining 8–13% improvements across demand levels. Q-Learning shows comparatively smaller but still large effects ($d = 0.66$ – 1.22), with EO producing the strongest separation ($d = 1.22$), yet limited transfer at 250 requests leads several configurations to exhibit deterioration (e.g., SEB: -10.66%). The gap in effect magnitudes between STGCRL ($d = 2.69$ – 3.99) and Q-Learning ($d = 0.66$ – 1.22) quantifies the divergent evaluation behavior and suggests that STGCRL's graph-based architecture more effectively captures spatial earnings regularities that generalize beyond the training regime, whereas Q-Learning struggles to maintain fairness gains outside training distributions, consistent with evidence that graph-based spatiotemporal models benefit from strong inductive bias, while standard RL methods exhibit limited generalization under distributional shift (Cobbe et al., 2019).



Figure 7. Earnings gini improvements achieved by fair agents across demand volumes

Workload fairness results (Figure 8) demonstrate the most consistent improvements across all experimental conditions, and normalized Cohen's d analysis confirms this metric exhibits the largest effect sizes of any performance measure. For STGCRL, SEB ($d = 4.40$) and B ($d = 4.07$) produce the strongest separations from the no-fairness baseline—the highest pairwise effects observed in the entire study—followed by UO ($d = 3.04$) and EO ($d = 1.93$), all exceeding the large-effect threshold. Q-Learning mirrors this pattern with SUB ($d = 2.80$) and B ($d = 2.01$) leading, and every configuration surpassing $|d| > 1.0$. Training achieves remarkable reductions at 750 requests, with all configurations delivering 21–40% improvements for both agents, and

evaluation confirms robust generalization with improvements ranging from 3–10% across most conditions. Notably, STGCRL consistently produces larger effect magnitudes than Q-Learning ($d = 1.68\text{--}4.40$ versus $1.10\text{--}2.80$), reinforcing the advantage of graph-based representations in capturing spatial utilization patterns. The similar performance across SEB, B, SUB, and UO configurations—all achieving large effects—validates that fairness interventions incorporating spatial and temporal context outperform static or purely income-based adjustments in mitigating workload disparities, particularly when allocation rules explicitly incorporate driver location and demand density (Jin et al., 2019; Lyu et al., 2025).

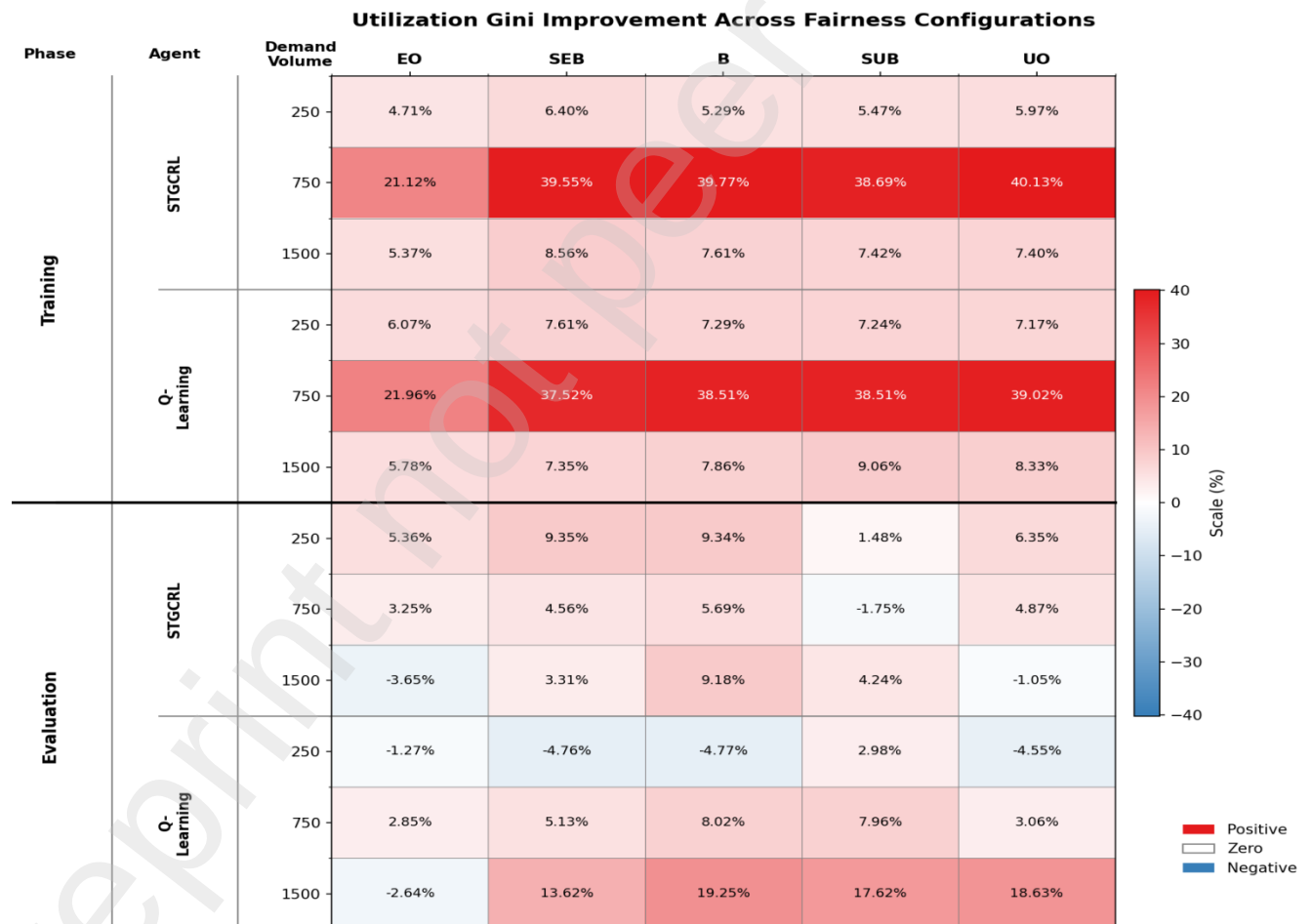


Figure 8. Utilization gini improvements achieved by fair agents across demand volumes

5. Conclusion

This study introduced GiniDispatch, a Gini-regularized, fairness-aware ride assignment algorithm for centralized taxi dispatch that jointly optimizes income and workload equity through position-based request prioritization, fairness-sensitive bipartite matching, and a composite reward function coupling trip-level returns with Gini-based fairness signals. Evaluated with STGCRL and tabular Q-Learning across three demand levels and six fairness configurations on the NYC Taxi dataset, GiniDispatch achieved substantial improvements in both earnings equality and workload equity. Cohen's d analysis revealed that workload and earnings equity produced the largest practical effects of any performance dimension, with Balanced and SEB configurations driving the strongest separations from the no-fairness baseline—suggesting that jointly targeting both equity dimensions captures complementary spatial regularities that single-objective alternatives miss. Utilization-oriented configurations similarly produced large effects, confirming that explicit workload redistribution serves as a powerful lever for overall fairness improvement. Critically, base reward effects remained negligible to small and waiting time showed no systematic pattern, demonstrating that fairness mechanisms operate without meaningful cost to platform revenue or passenger experience. Trip efficiency improved under every fairness configuration, revealing that equitable driver distribution is not merely a social objective but an operational advantage that reduces idle repositioning and spatial clustering. STGCRL consistently produced larger effect magnitudes than Q-Learning across all fairness-sensitive metrics, indicating that graph-based spatiotemporal representations amplify the benefits of fairness-aware dispatch by capturing spatial dependencies that tabular methods cannot exploit. These findings establish that fairness constraints in ride-hailing dispatch need not be framed as a trade-off against efficiency, and that both the

fairness configuration and learning architecture substantially shape the magnitude of achievable equity gains.

A central finding of this work is the critical importance of a multidimensional approach to fairness in taxi dispatch. Configurations that focused exclusively on earnings equality inadvertently eroded overall driver welfare, underscoring that income parity alone is an insufficient proxy for equitable treatment—a pattern corroborated by the Cohen's d analysis, where earnings-only configurations produced notably weaker effect sizes across workload and efficiency metrics compared to configurations that jointly targeted both equity dimensions. By explicitly integrating workload equality alongside earnings equality, GiniDispatch addresses a dimension of driver welfare that has received comparatively little attention in the fairness-aware dispatch literature (Sühr et al., 2019; Liang, 2024). The consistently large effect sizes observed under Balanced, SEB, and utilization-oriented configurations suggest that workload redistribution activates complementary fairness mechanisms that earnings-focused interventions alone cannot reach, reinforcing the view that platform workers' well-being is shaped not only by compensation levels but also by the duration and intensity of labor (Allon et al., 2023; Göktaş, 2023; Selcuk & Gokpinar, 2025). In practice, drivers who must work significantly longer hours to achieve comparable earnings face heightened fatigue, elevated accident risk, and diminished job satisfaction (Lim & Chia, 2015; Meng et al., 2015), all of which can trigger increased turnover and ride cancellations that ultimately degrade system performance (Zhou et al., 2024). That configurations incorporating utilization equality also improved trip efficiency further demonstrates that workload-aware dispatch generates system-wide operational benefits beyond the equity objective itself. GiniDispatch's ability to mitigate both earnings and workload disparities simultaneously thus addresses a structural gap in current dispatch optimization, and the effect size evidence makes clear that the

choice of fairness configuration is not a secondary design decision but a primary determinant of how deeply equity gains penetrate across performance dimensions.

The practical implications of this study extend to platform operators, regulators, and policymakers concerned with the social welfare of gig economy workers. The Cohen's d analysis provides quantitative evidence that centralized dispatch systems can achieve negligible effect sizes on base reward and waiting time while producing large effects on earnings and workload equity—demonstrating that fairness constraints operate as a near-costless intervention rather than a performance trade-off. This finding is particularly significant given the growing regulatory and public pressure on ride-hailing platforms to address driver income inequality and working conditions (Dang et al., 2025; Chen et al., 2025). Furthermore, the demonstrated effectiveness of even a relatively simple RL agent (Q-Learning) in producing large fairness effect sizes when guided by well-designed reward shaping—though consistently smaller in magnitude than STGCRL—suggests that the algorithmic infrastructure required for equitable dispatch need not be prohibitively complex, an important consideration for smaller platforms or those operating in emerging markets where computational resources may be constrained. The effect size gap between the two architectures does, however, indicate that platforms with access to graph-based spatiotemporal modeling stand to extract substantially deeper equity gains from the same fairness configurations, offering a clear upgrade path as operational capacity grows.

This study has several limitations that warrant acknowledgment. First, the evaluation was conducted using data from a single metropolitan area (New York City), which may limit the generalizability of findings to cities with fundamentally different spatial configurations, demand patterns, or mobility infrastructures. Recent work on multi-city dispatch evaluations has shown that fairness-efficiency trade-offs can vary substantially across urban geographies (Jiang et al.,

2025; Sun et al., 2022), highlighting the need for cross-city validation. Second, although three demand levels were analyzed, a more granular temporal sampling strategy could better capture intraday fluctuations in travel behavior and their interaction with fairness dynamics. Third, the focus on value-based (STGCRL) and tabular (Q-Learning) RL paradigms, while deliberate for establishing baseline comparisons, constrained the inclusion of policy-gradient or actor-critic methods (e.g., A2C, PPO) that may offer complementary advantages in handling the fairness–efficiency trade-off through constrained optimization formulations. Fourth, the fleet size of 250 taxis, while appropriate for the dataset characteristics and aligned with prior studies, represents a single configuration; future work should conduct sensitivity analyses across varying fleet sizes (e.g., 100, 500, 1000 taxis) to examine how fairness mechanisms scale with fleet density. Finally, the assumption of full driver compliance with dispatch assignments overlooks the behavioral realism of real-world operations, where drivers may reject unfavorable assignments due to congestion, opportunity costs, or contextual preferences (Selcuk & Gokpinar, 2025; Zhou et al., 2024).

Several promising directions emerge for future research. First, extending GiniDispatch to multi-agent settings, where fairness emerges from the collective behavior of decentralized agents coordinating through shared fairness signals, could enhance both scalability and robustness in large fleet deployments (Rajeh et al., 2024; Wang et al., 2025). Second, exploring hierarchical or region-level fairness models—where neighborhood-level equity constraints interact with system-wide efficiency objectives—would improve both interpretability and policy relevance for urban planners. Third, incorporating behavioral realism, including partial compliance modeling, learning-induced fatigue, and strategic rejection behavior, would bridge the gap between simulation and real-world dispatch operations. Fourth, integrating passenger-side fairness

considerations, such as waiting time equity across geographic regions (Jiang et al., 2025), with the driver-side equity mechanisms proposed here would move toward a holistic, multi-stakeholder fairness framework for ride-hailing platforms. Finally, benchmarking GiniDispatch against hybrid optimization frameworks that combine heuristic matching with adaptive learning, as well as emerging approaches leveraging large language models for dispatch decision-making (Lyu et al., 2025), may offer valuable insights into the scalability and adaptability of fairness-oriented dispatch systems in increasingly complex urban environments.

Statements/ Declarations

Declaration of Interest Statement

The authors have no competing interests to declare that are relevant to the content of this article.

Data Availability

The authors have shared their detailed results, scripts, and the datasets.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) utilized ChatGPT to enhance the language of the current paper, and Claude Opus 4.5. for Code Debugging. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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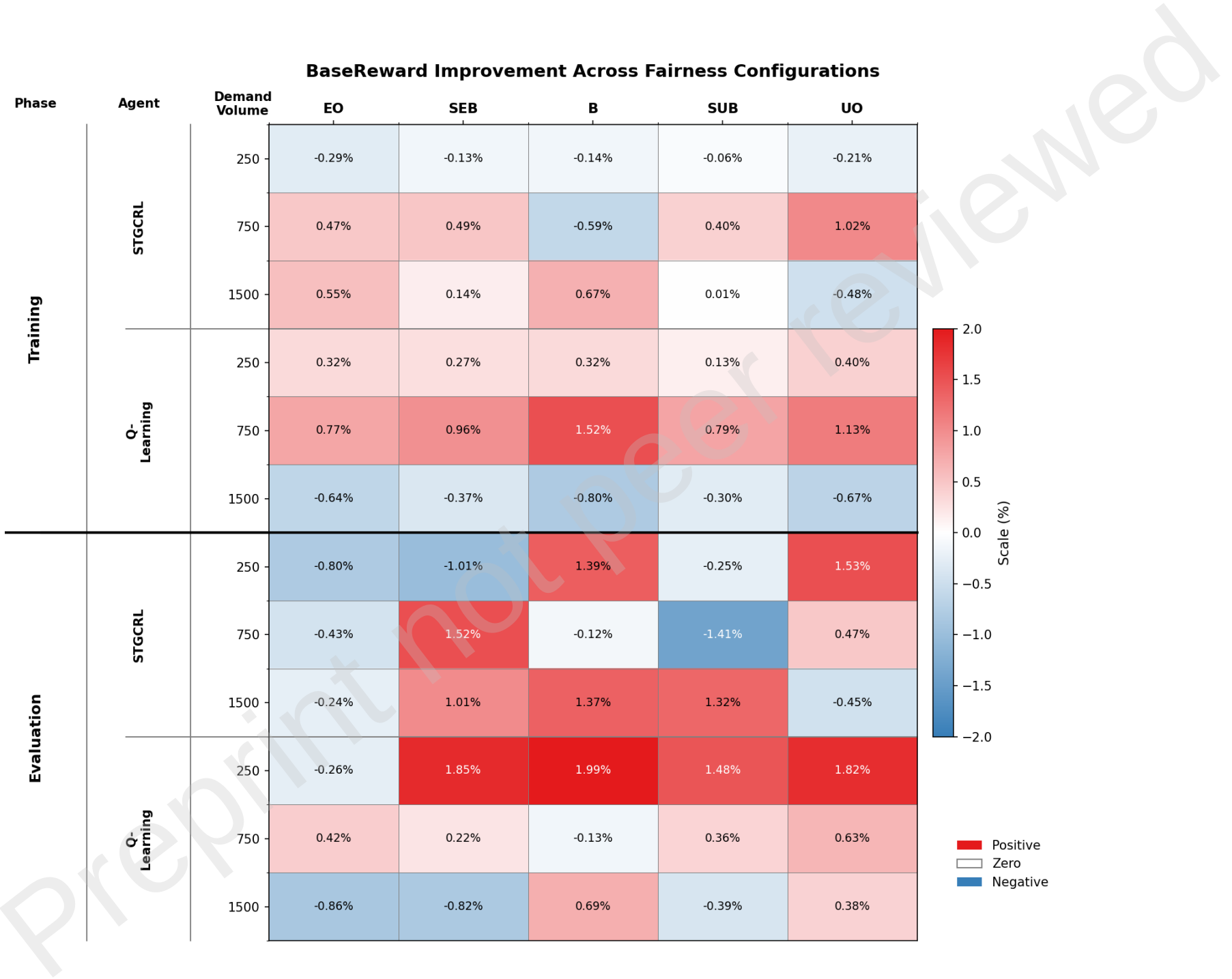
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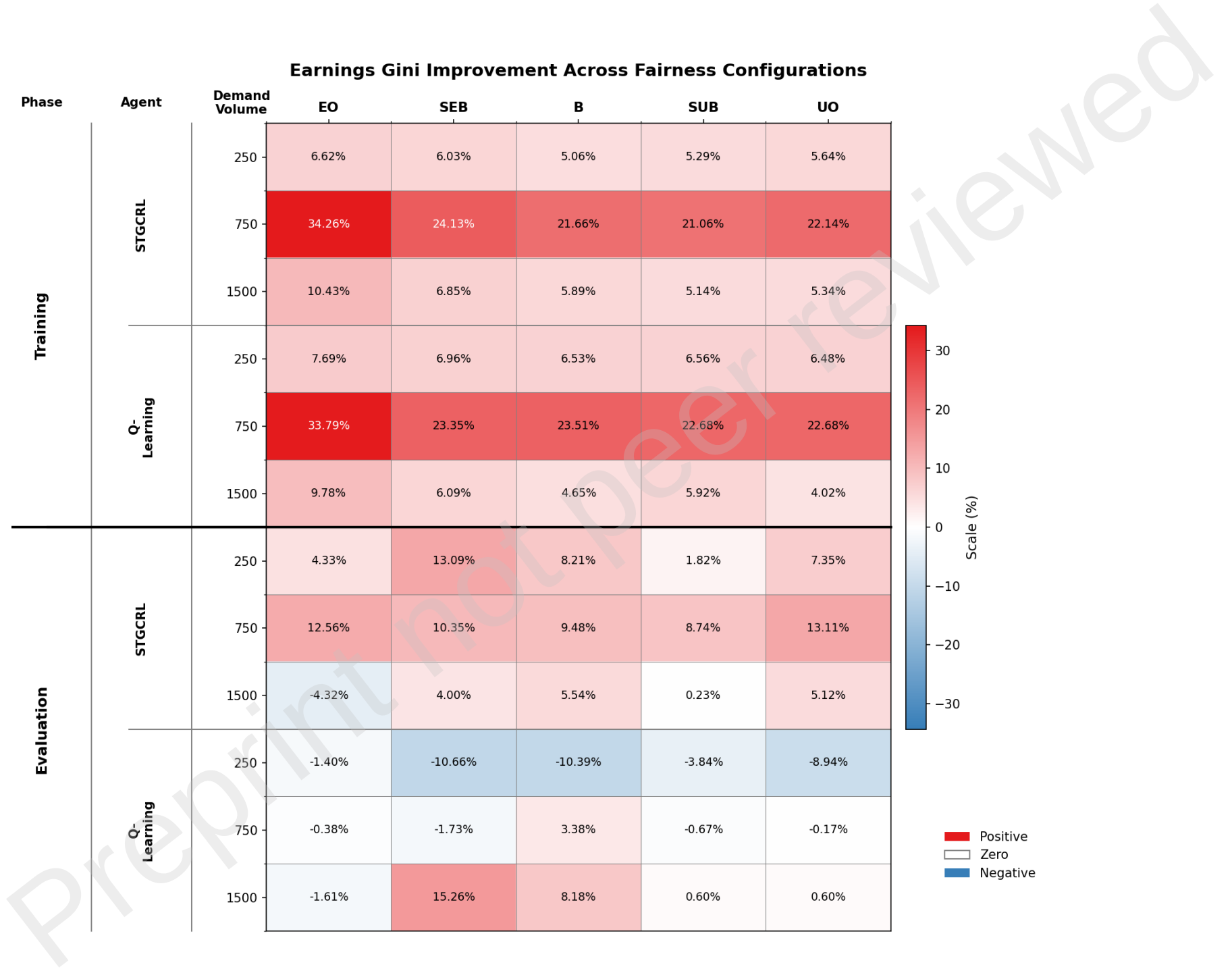
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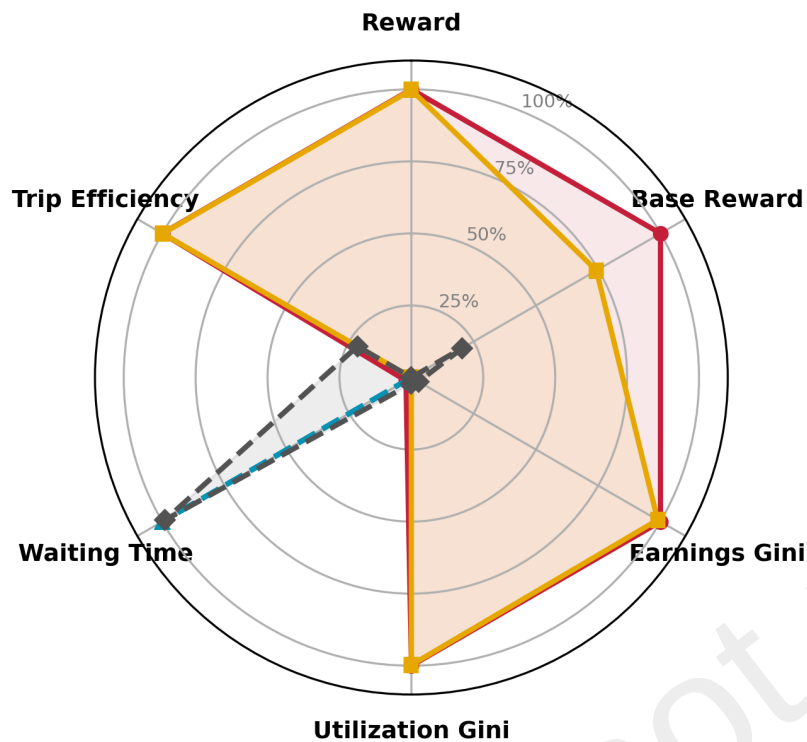
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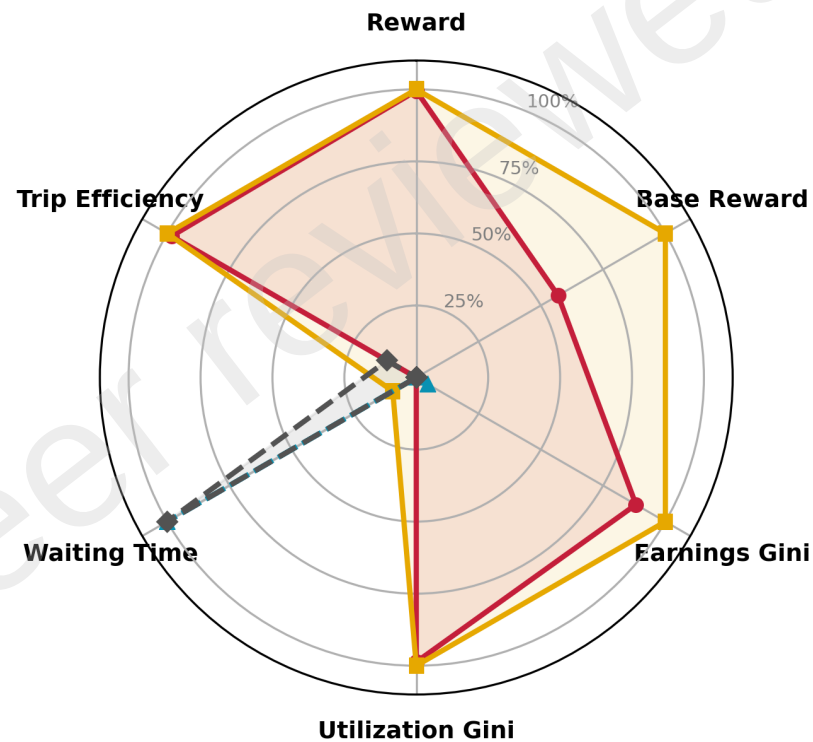




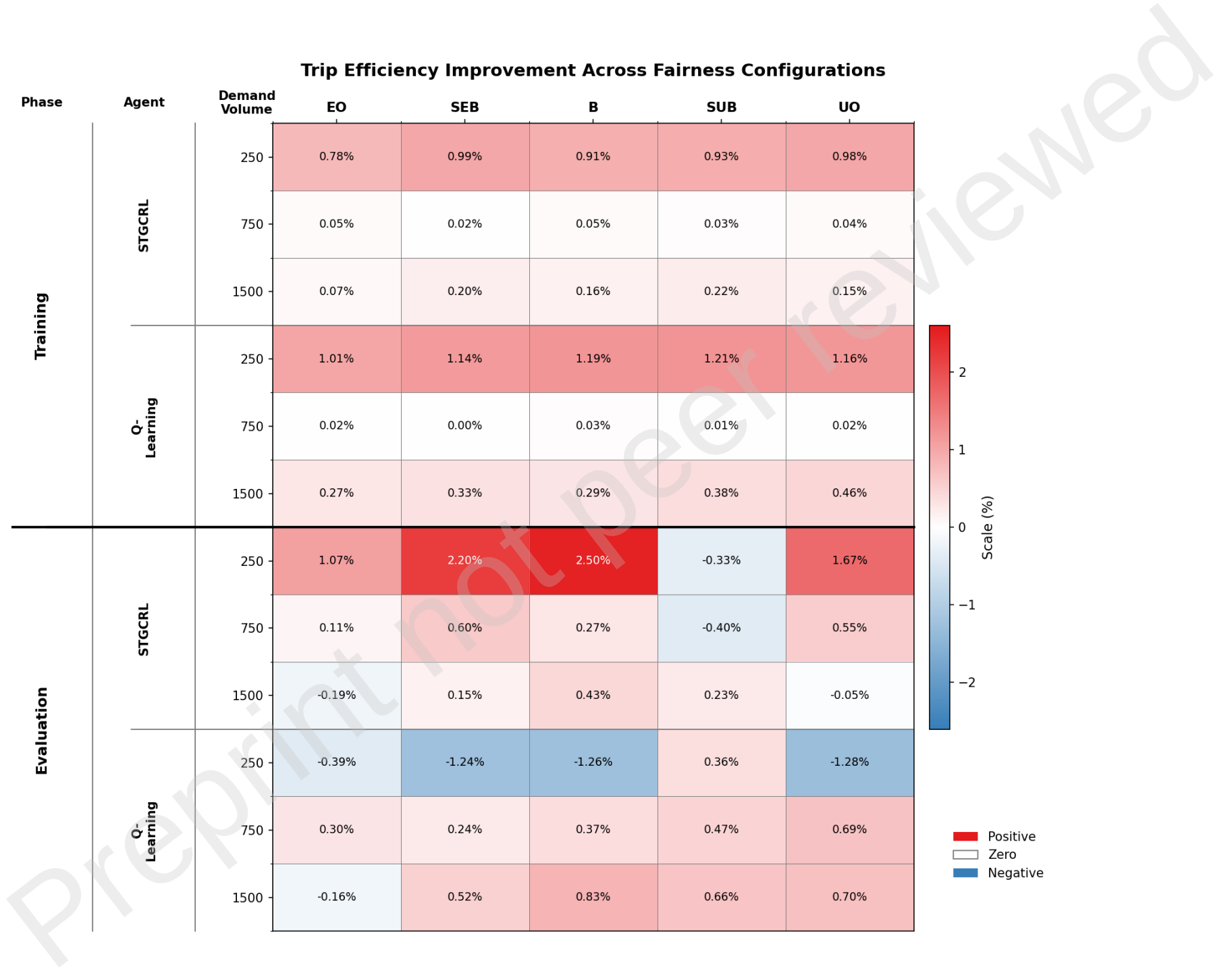
Train Metrics

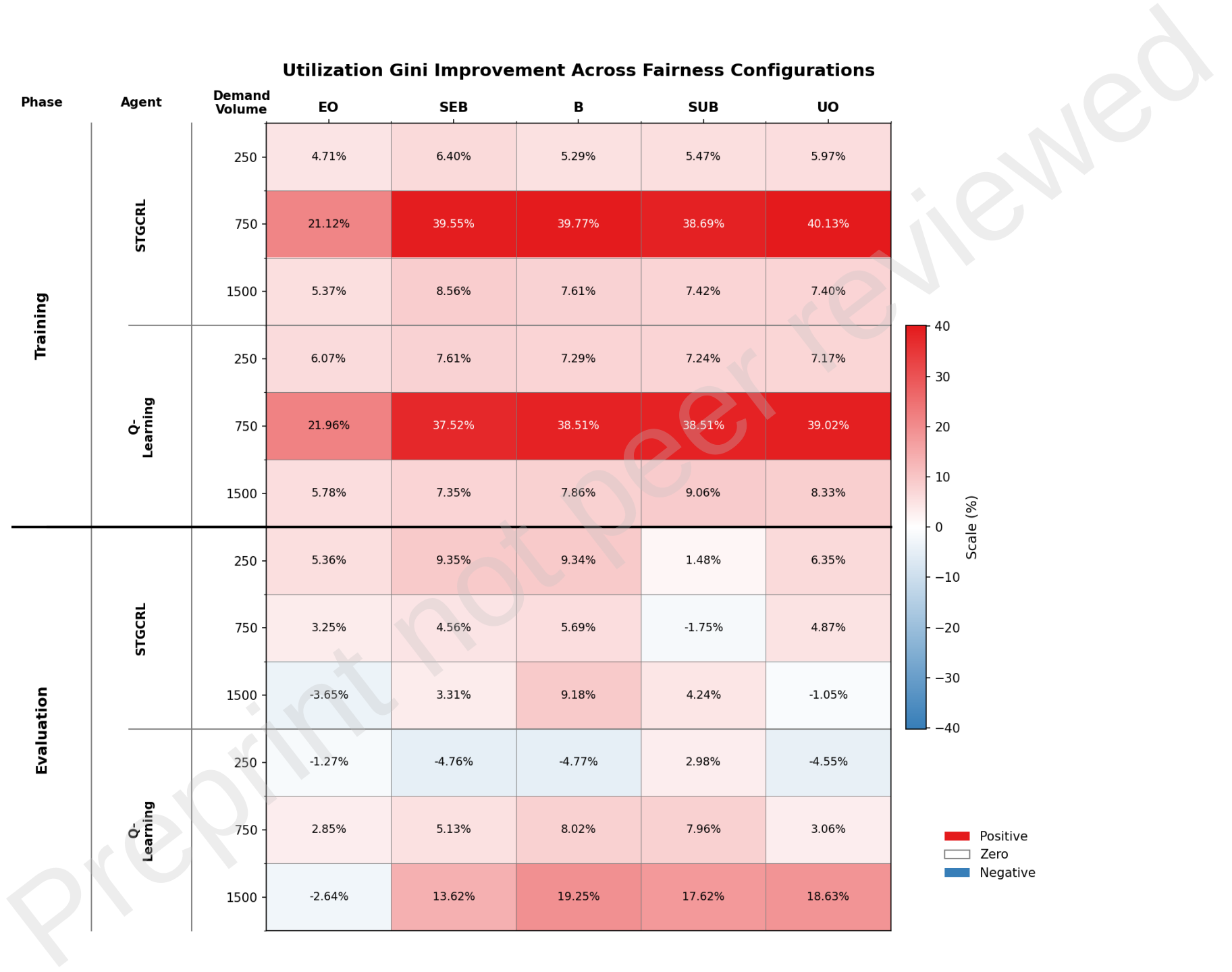


Evaluate Metrics



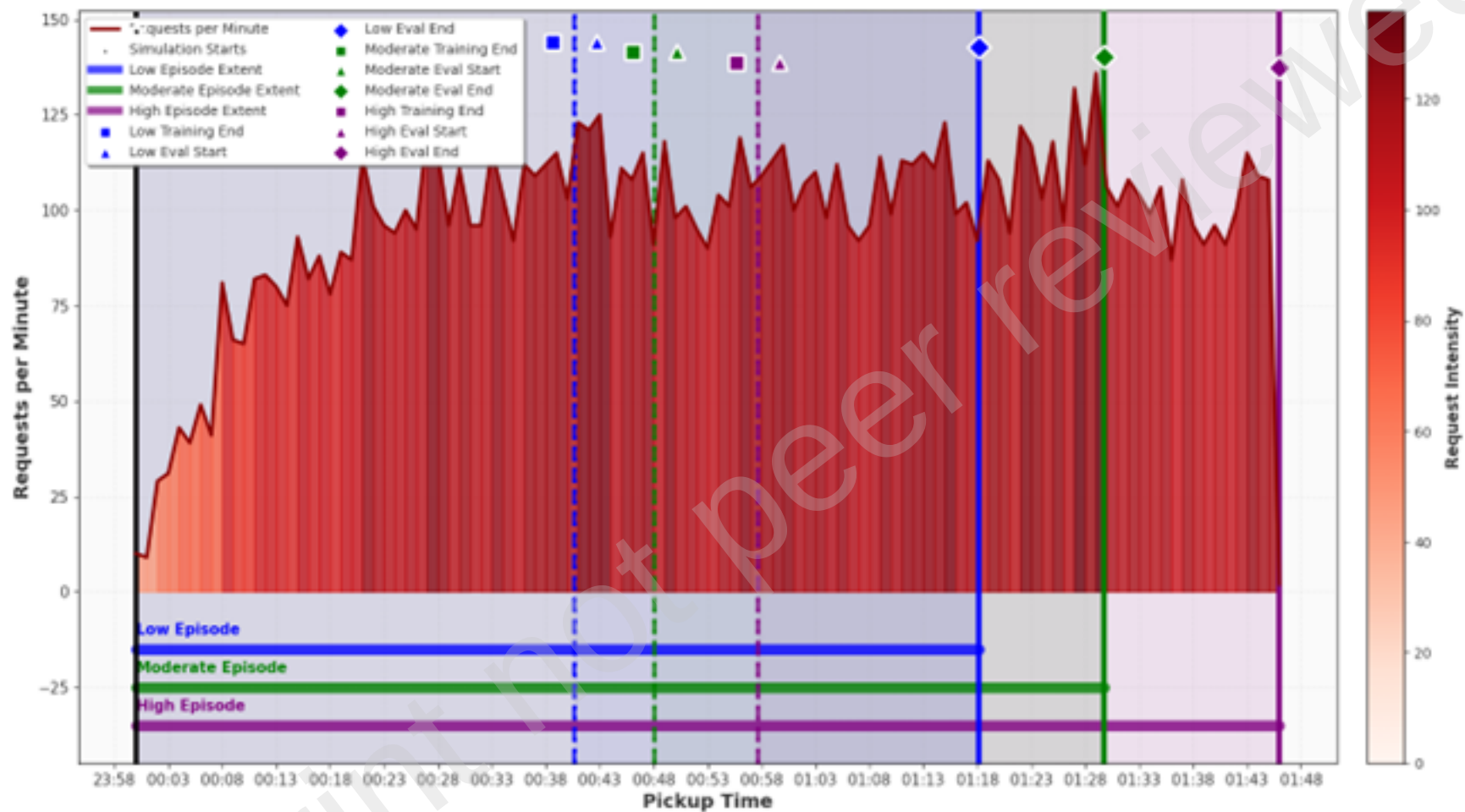
STGCRL Q-Learning Nearest Greedy Hungarian Algorithm







Taxi Request Distribution Across Demand Levels



Total Reward Improvement

